

Notes for your Keras3/Tensorflow/pytorch environnement

Hands-on 1 Classification



— Montreal, June 15–19 —

DLMI2026

Preliminary remarks

- All four hands-on can be done on a computing cloud such as that of *CalculQuebec*, *Google Colab*, ... or on your own computer
- **For beginners** : we recommend you to start with *CalculQuebec*
- For advanced users with CPU/GPU access
 - You can run on your specific hardware
 - Tips for installing the required environnement are [here](#).

First, setup your *CalculQuebec* Account

(cf. DLMI 2026 hands on setup instructions)

1. *Create an account on CalculQuebec Cloud and log-in*
<https://jupyter.dmlri.calculquebec.cloud/hub/login>
2. Create a compute node with
 1. Memory at 4096MB
 2. GPU 1x1G.5GB
3. Start the jupyter-lab server and wait for it **Steps covered here**
4. Download the hands-on material into your jupyter-lab
5. Open a notebook
6. Cells philosophy and execution

Jupyter Lab interface (can change)

The screenshot shows the Jupyter Lab web interface. On the left is a file browser with a table of files and folders. In the center is a 'Launcher' tab displaying various application icons. On the right is a 'Terminal' section with icons for different environments. Annotations include orange arrows pointing to the file browser, a green arrow pointing to the 'Launcher' tab, and another green arrow pointing to the 'Terminal' section. A red text box is overlaid on the file browser area.

File Edit View Run Kernel Tabs Settings Help

← → ↻ 🔍 jupyter.dml.culquebec.cloud/user/pmjodoin/lab

Launcher

Notebook

Python 3.11 LibreQDA [?] Mate Desktop (VNC) [?] MLflow [?] OpenRefine [?] Python bundle RStudio [?] Tensorboard [?] VS Code [?]

XFCE4 Desktop (VNC) [?]

Console

Python 3.11 Python bundle

Other

Terminal Job statistics Text File Markdown File Python File Show Contextual Help

Documentation

General Jupyter

Would you like Jupyter news?

Drag and drop files here (or via the upload button)

Launcher tab

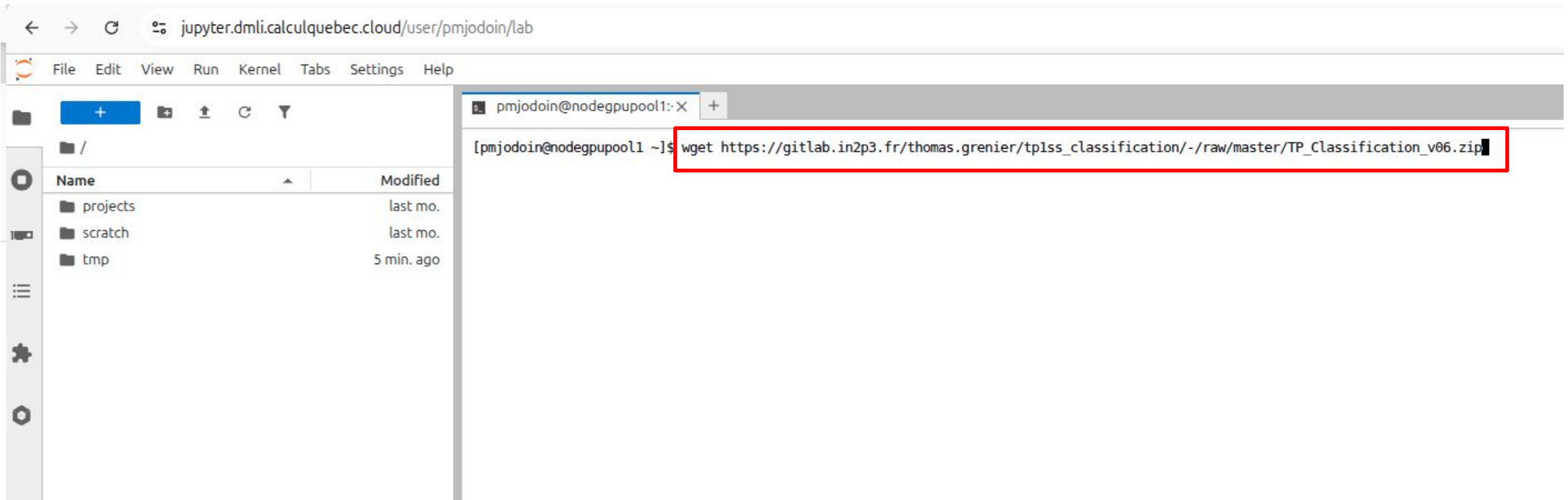
Terminal

Download Hands-on materials

a- Open a **Terminal** (c.f. previous page)

b- download from the terminal with the following command :

```
wget https://gitlab.in2p3.fr/thomas.grenier/tp1ss_classification/-/raw/master/TP_Classification_v06.zip
```



Download Hands-on materials

a- Open a **Terminal**

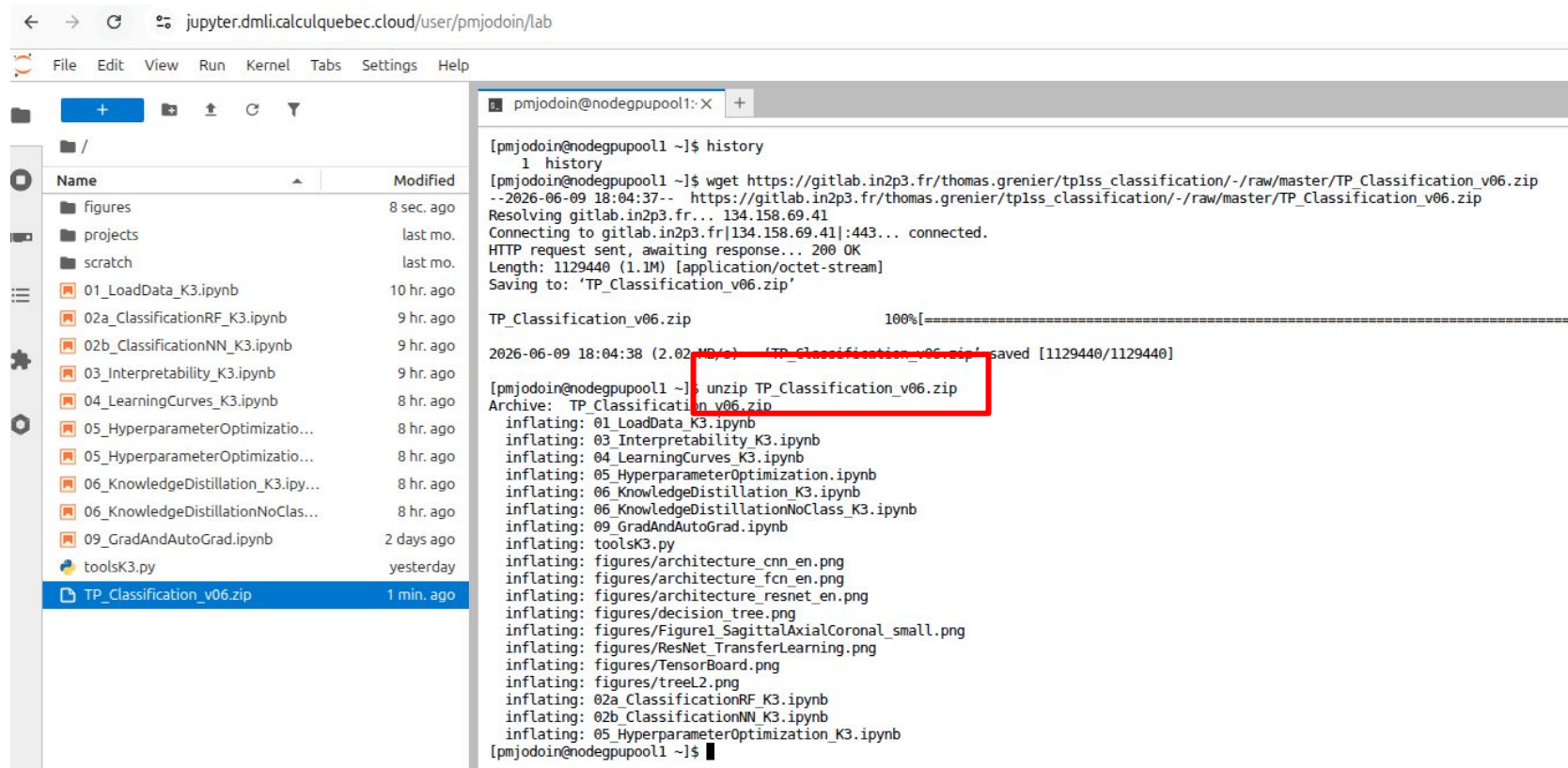
b- download from the terminal with :

```
wget https://gitlab.in2p3.fr/thomas.grenier/tp1ss_classification/-/raw/master/TP_Classification_v06.zip
```

c- unzip the file (in a Terminal)

```
unzip TP_Classification_v06.zip
```

unzipped files



```
[pmjodoin@nodegpupool1 ~]$ history
1 history
[pmjodoin@nodegpupool1 ~]$ wget https://gitlab.in2p3.fr/thomas.grenier/tp1ss_classification/-/raw/master/TP_Classification_v06.zip
--2026-06-09 18:04:37-- https://gitlab.in2p3.fr/thomas.grenier/tp1ss_classification/-/raw/master/TP_Classification_v06.zip
Resolving gitlab.in2p3.fr... 134.158.69.41
Connecting to gitlab.in2p3.fr[134.158.69.41]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1129440 (1.1M) [application/octet-stream]
Saving to: 'TP_Classification_v06.zip'

TP_Classification_v06.zip      100%[=====] 1129440/1129440
2026-06-09 18:04:38 (2.02 MB/s) 'TP_Classification_v06.zip' saved [1129440/1129440]

[pmjodoin@nodegpupool1 ~]$ unzip TP_Classification_v06.zip
Archive: TP_Classification_v06.zip
  inflating: 01_LoadData_K3.ipynb
  inflating: 03_Interpretability_K3.ipynb
  inflating: 04_LearningCurves_K3.ipynb
  inflating: 05_HyperparameterOptimization.ipynb
  inflating: 06_KnowledgeDistillation_K3.ipynb
  inflating: 06_KnowledgeDistillationNoClass_K3.ipynb
  inflating: 09_GradAndAutoGrad.ipynb
  inflating: toolsK3.py
  inflating: figures/architecture_cnn_en.png
  inflating: figures/architecture_fcnn_en.png
  inflating: figures/architecture_resnet_en.png
  inflating: figures/decision_tree.png
  inflating: figures/Figure1_SagittalAxialCoronal_small.png
  inflating: figures/ResNet_TransferLearning.png
  inflating: figures/TensorBoard.png
  inflating: figures/treeL2.png
  inflating: 02a_ClassificationRF_K3.ipynb
  inflating: 02b_ClassificationNN_K3.ipynb
  inflating: 05_HyperparameterOptimization_K3.ipynb
[pmjodoin@nodegpupool1 ~]$
```

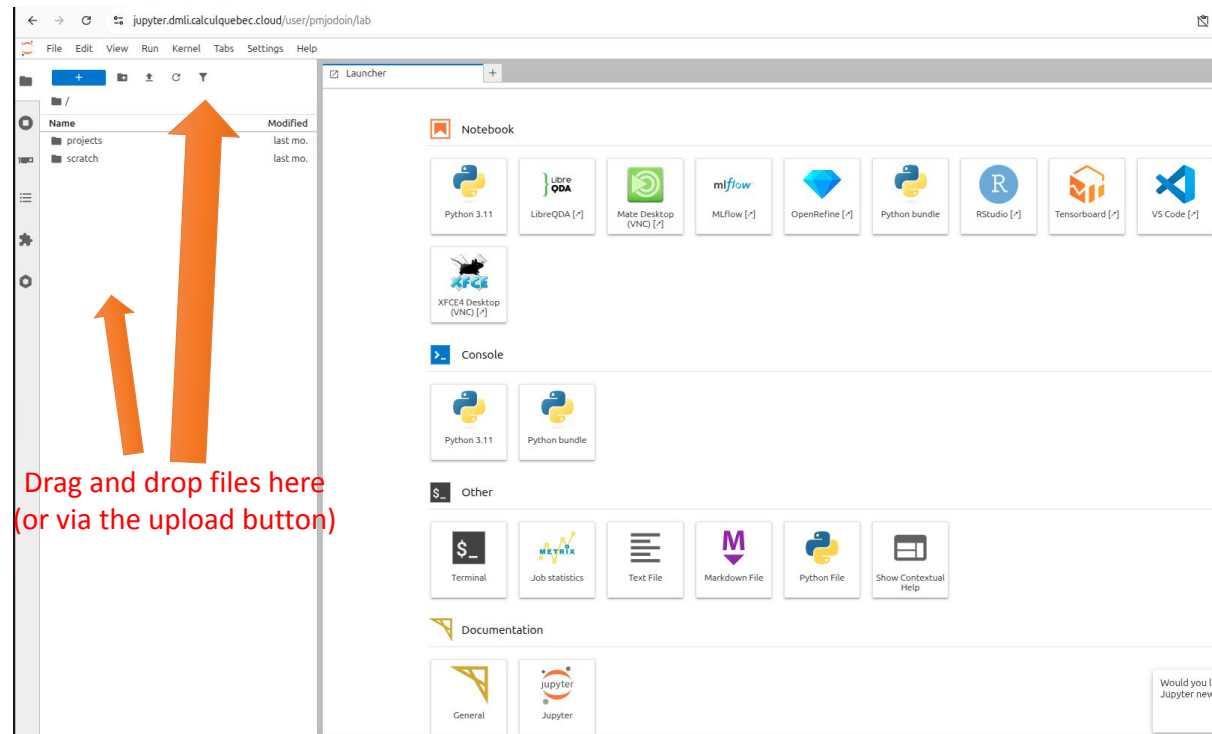
Download Hands-on materials

If the previous step did not work, you can

a) download the zip file on your computer

https://gitlab.in2p3.fr/thomas.grenier/tp1ss_classification/-/raw/master/TP_Classification_v06.zip

b) Upload it in your online environment



Download Hands-on materials

If the previous step did not work, you can

a) download the zip file on your computer

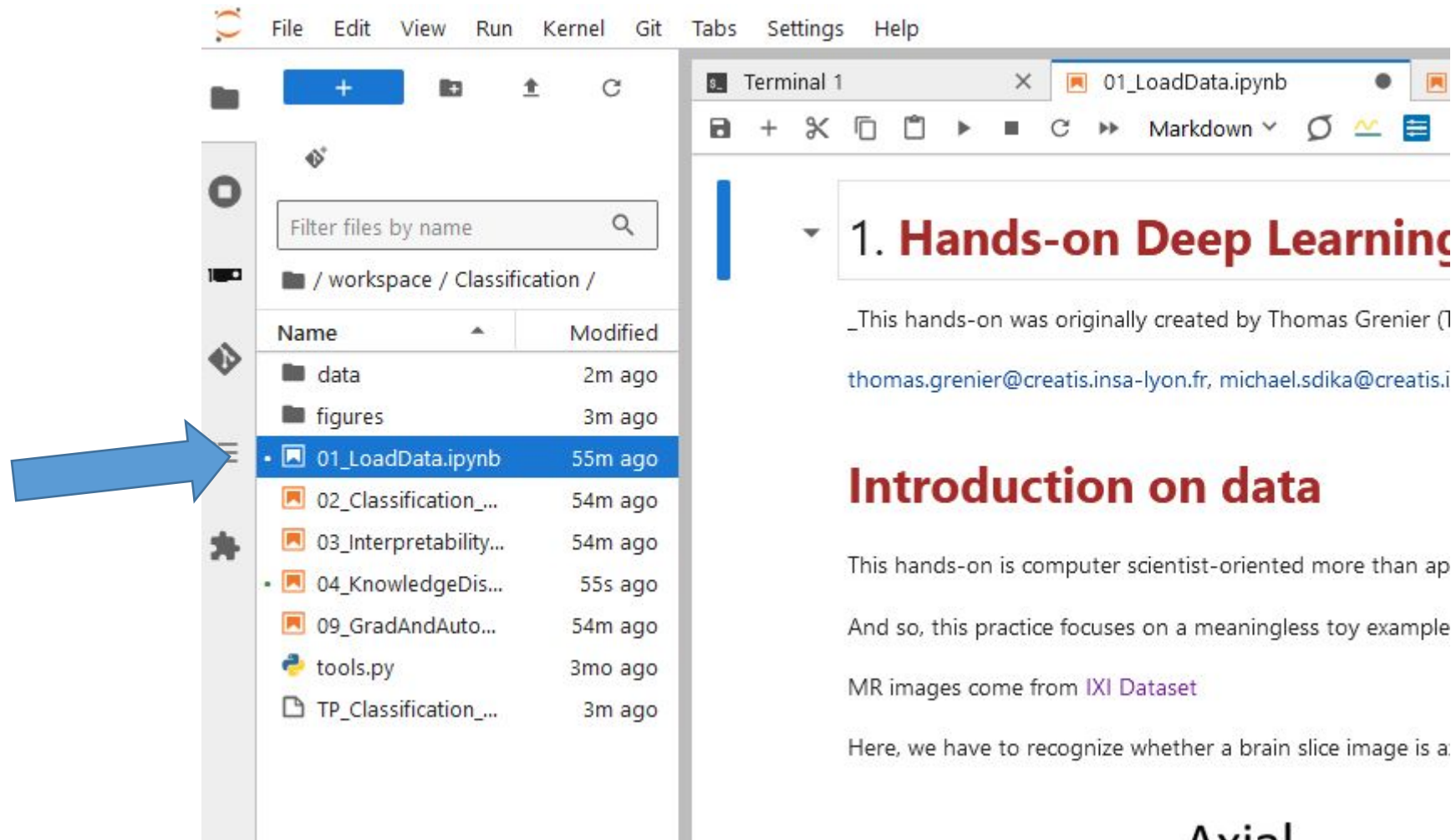
https://gitlab.in2p3.fr/thomas.grenier/tp1ss_classification/-/raw/master/TP_Classification_v06.zip

b) Upload it in your online environment

c) Unzip the file

Select and open the desired notebook

As example, double clic on : 01_LoadData.ipynb

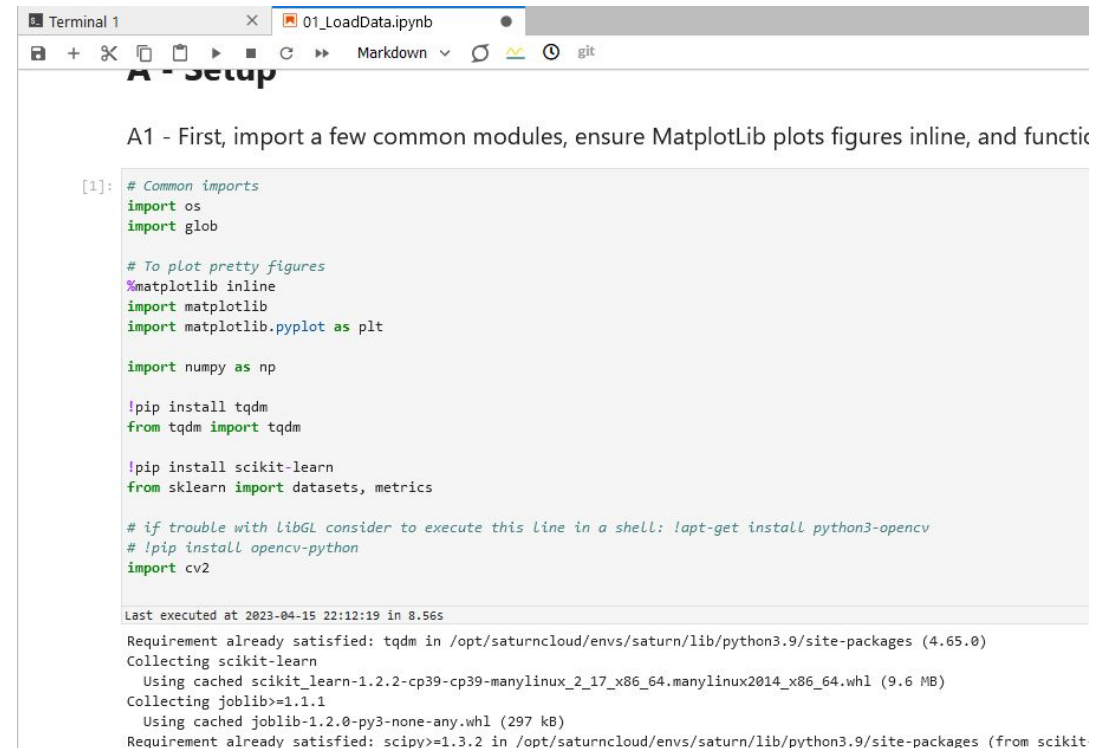


The screenshot displays the JupyterLab interface. On the left, the file browser shows the directory structure: / workspace / Classification /. A list of files and folders is shown, including 'data', 'figures', '01_LoadData.ipynb', '02_Classification_...', '03_Interpretability...', '04_KnowledgeDis...', '09_GradAndAuto...', 'tools.py', and 'TP_Classification_...'. A blue arrow points to the '01_LoadData.ipynb' file, which is highlighted. The main area on the right shows the notebook content, titled '1. Hands-on Deep Learning'. The text in the notebook includes: '_This hands-on was originally created by Thomas Grenier (1 thomas.grenier@creatis.insa-lyon.fr, michael.sdika@creatis.i Introduction on data This hands-on is computer scientist-oriented more than ap And so, this practice focuses on a meaningless toy example MR images come from IXI Dataset Here, we have to recognize whether a brain slice image is a.

Avial

Cells and running code

- Notebooks are made of cells : they can contain text or python code
- Python code can be executed from the jupyter lab with the play button 
- Outputs are added in a new cell after the code executed
- A cell can be executed many times
- Generally, a cell requires that the preceding cells have been executed before



The screenshot shows a Jupyter Lab window with a terminal and a code editor. The code editor displays a Python cell with the following content:

```
A1 - First, import a few common modules, ensure Matplotlib plots figures inline, and function
```

```
[1]: # Common imports
import os
import glob

# To plot pretty figures
%matplotlib inline
import matplotlib
import matplotlib.pyplot as plt

import numpy as np

!pip install tqdm
from tqdm import tqdm

!pip install scikit-learn
from sklearn import datasets, metrics

# if trouble with LibGL consider to execute this line in a shell: !apt-get install python3-opencv
# !pip install opencv-python
import cv2
```

Below the code, the execution output is shown:

```
Last executed at 2023-04-15 22:12:19 in 8.56s
Requirement already satisfied: tqdm in /opt/saturncloud/envs/saturn/lib/python3.9/site-packages (4.65.0)
Collecting scikit-learn
  Using cached scikit_learn-1.2.2-cp39-cp39-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (9.6 MB)
Collecting joblib>=1.1.1
  Using cached joblib-1.2.0-py3-none-any.whl (297 kB)
Requirement already satisfied: scipy>=1.3.2 in /opt/saturncloud/envs/saturn/lib/python3.9/site-packages (from scikit
```

Hands on 1

For Hands on 1, please focus on the first three Notebooks

1. Hands on 1

2. Introduction

Axial Sagittal Coronal

T1